BRAKE LAMP SWITCH (Continued)

OPERATION

Vehicles equipped with the speed control option use a multiple function brake lamp switch. The PCM monitors the state of the brake lamp switch. Refer to the Brake section for more information on brake lamp switch service and adjustment procedures.

The primary function of the brake switch is to turn on the brake lamps during braking. The switch is also used to send signals to components that must know when the brakes are applied, such as the Powertrain Control Module (PCM), which uses the signal to cancel speed control. The Controller Antilock Brake (CAB) uses the brake switch signal to monitor brake pedal application. When the normally closed switch contacts open, the CAB receives the brake applied signal. The CAB then monitors the ABS system to anticipate the need for a ABS stop.

DIAGNOSIS AND TESTING - BRAKE LAMP SWITCH

The brake lamp switch operation can be tested with an ohmmeter. The ohmmeter is used to check continuity between the pin terminals (Fig. 1).

SWITCH CIRCUIT IDENTIFICATION

- Terminals 1 and 2: brake sensor circuit
- Terminals 3 and 4: speed control circuit if equipped
- Terminals 5 and 6: brake lamp circuit

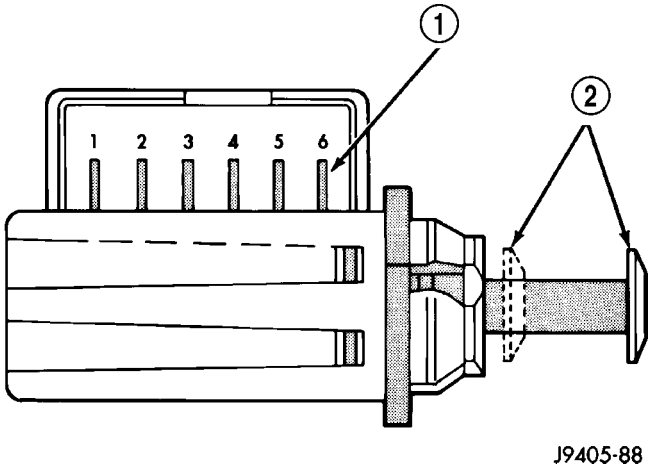

Fig. 1 Brake Lamp Switch Terminal Identification

1 - TERMINAL PINS
2 - PLUNGER TEST POSITIONS

SWITCH CONTINUITY TEST

NOTE: Disconnect switch harness before testing switch continuity.

With the switch plunger retracted, attach test leads to terminal pins 1 and 2. Replace switch if meter indicates no continuity.

With the switch plunger retracted, attach test leads to terminal pins 3 and 4. Replace switch if meter indicates no continuity.

With the switch plunger extended, attach test leads to terminal pins 5 and 6. Replace switch if meter indicates no continuity.

REMOVAL

(1) Remove the steering column cover and the lower trim panel.

(2) Press the brake pedal downward to fully applied position.

(3) Rotate the switch approximately 30° in counterclockwise direction. Then pull the switch rearward and out of bracket.

(4) Disconnect the switch harness and remove the switch (Fig. 2).

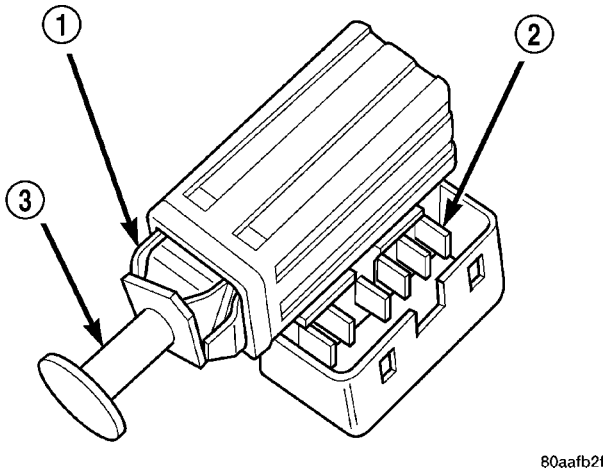

Fig. 2 Brake Lamp Switch

1 - RETAINER
2 - TERMINALS
3 - SWITCH PLUNGER

INSTALLATION

(1) Pull the switch plunger all of the way out, to fully extended position.

(2) Connect the harness connector to the switch.

(3) Press and hold the brake pedal in the applied position.

(4) Align the tab on the switch with the notch in the switch bracket. Then insert the switch in the bracket and turn it clockwise about 30° to lock it in place.

(5) Release the brake pedal, then pull the pedal fully rearward. Pedal will set the plunger to the correct position as the pedal pushes the plunger into the switch body. The switch will make a ratcheting sound as it self-adjusts.